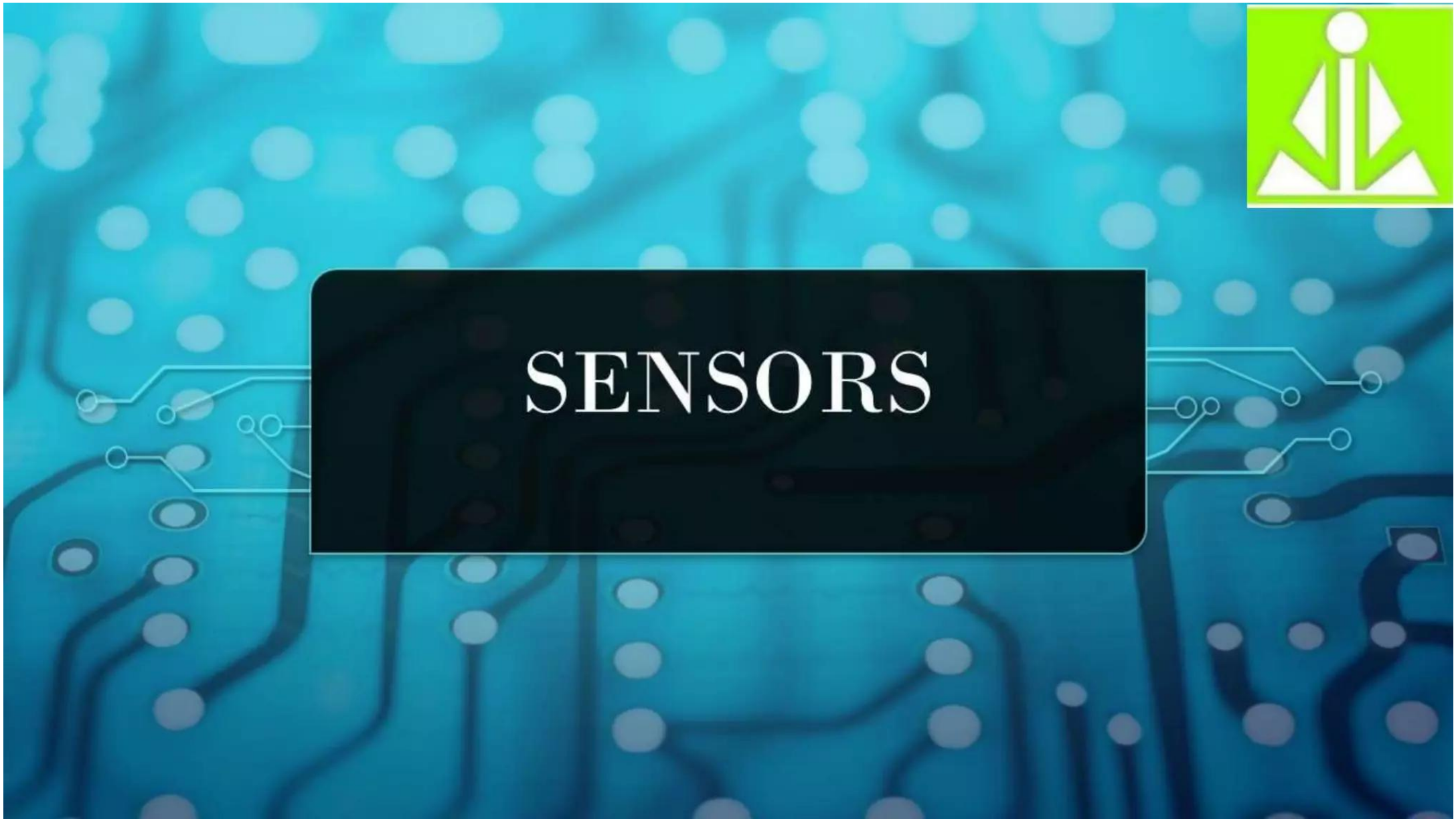

SENSORS

WHAT IS A SENSOR?

- Sensor as an input device which provides an output (signal) with respect to a specific physical quantity (input).
- The term “input device” in the definition of a Sensor means that it is part of a bigger system which provides input to the main control system (like a Processor or a Microcontroller).
- Another unique definition of a Sensor is as follows: It is a device that converts signals from one energy domain to the electrical domain.

CLASSIFICATION OF SENSORS

In the first classification of the sensors, they are divided in to Active and Passive.

- **Active Sensors** are those which require an external *citation* signal or a power signal.
- **Passive Sensors**, on the other hand, do not require any external power signal and directly generate output response.

The final classification of the sensors is **Analog and Digital Sensors.**

- **Analog Sensors** produce an analog output i.e. a continuous output signal with respect to the quantity being measured.
- **Digital Sensors,** in contrast to Analog Sensors, work with discrete or digital data. The data in digital sensors, which are used for conversion and transmission, is digital in nature.

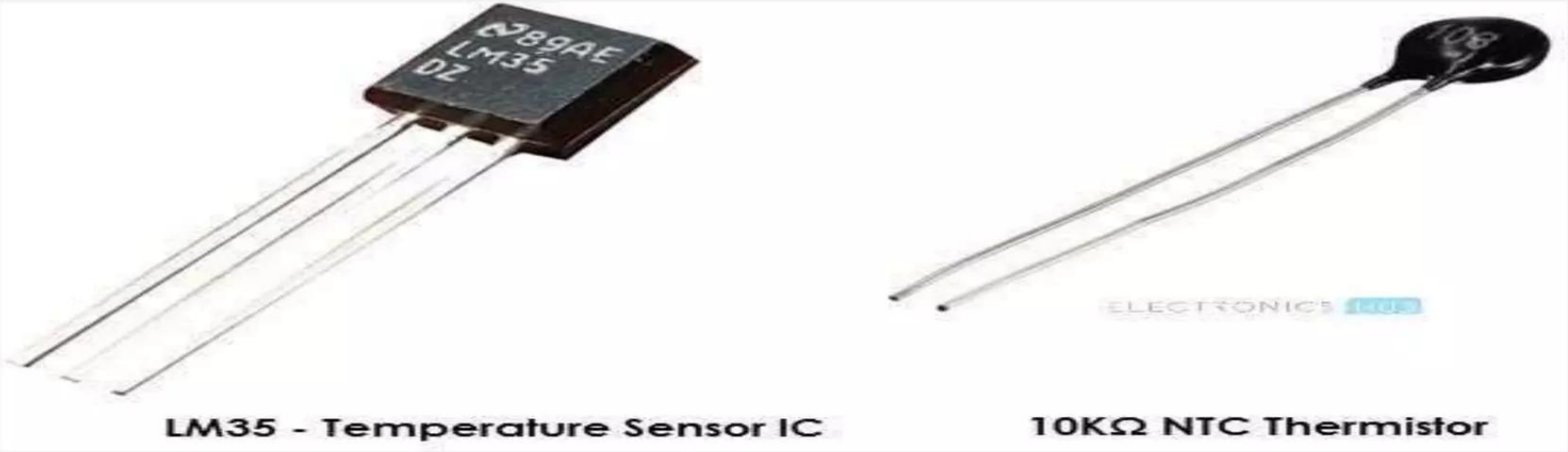
DIFFERENT TYPES OF SENSORS

All these sensors are used for measuring one of the physical properties like Temperature, Resistance, Capacitance, Conduction, Heat Transfer etc.

- Temperature Sensor
- Proximity Sensor
- Accelerometer
- IR Sensor (Infrared Sensor)
- Pressure Sensor
- Light Sensor
- Ultrasonic Sensor
- Smoke, Gas and Alcohol Sensor
- Touch Sensor
- Color Sensor
- Humidity Sensor
- Tilt Sensor
- Flow and Level Sensor

TEMPERATURE SENSOR

- One of the most common and most popular sensors is the Temperature Sensor. A Temperature Sensor, as the name suggests, senses the temperature i.e. it measures the changes in the temperature. In a temperature sensor, the temperature changes correspond to **changes in its physical property like resistance or voltage.**
- **Thermistor**
- **Thermocouple**
- A **thermistor** can be used to ***detect the variation in temperature*.** It has a negative temperature coefficient that means when the temperature increases the resistance decreases. So, the thermistor's resistance can be varied with the rise in temperature which causes more current flow through it.



- Another component that can ***detect the variation in temperature*** **is a thermocouple.** In its construction, two different metals are joined together to form a junction
- **LM35** is a Celsius scale temperature sensor device with its output directly proportional to the temperature. LM35 can measure temperatures in the range of -55^0C to $+150^0$C

PROXIMITY SENSORS

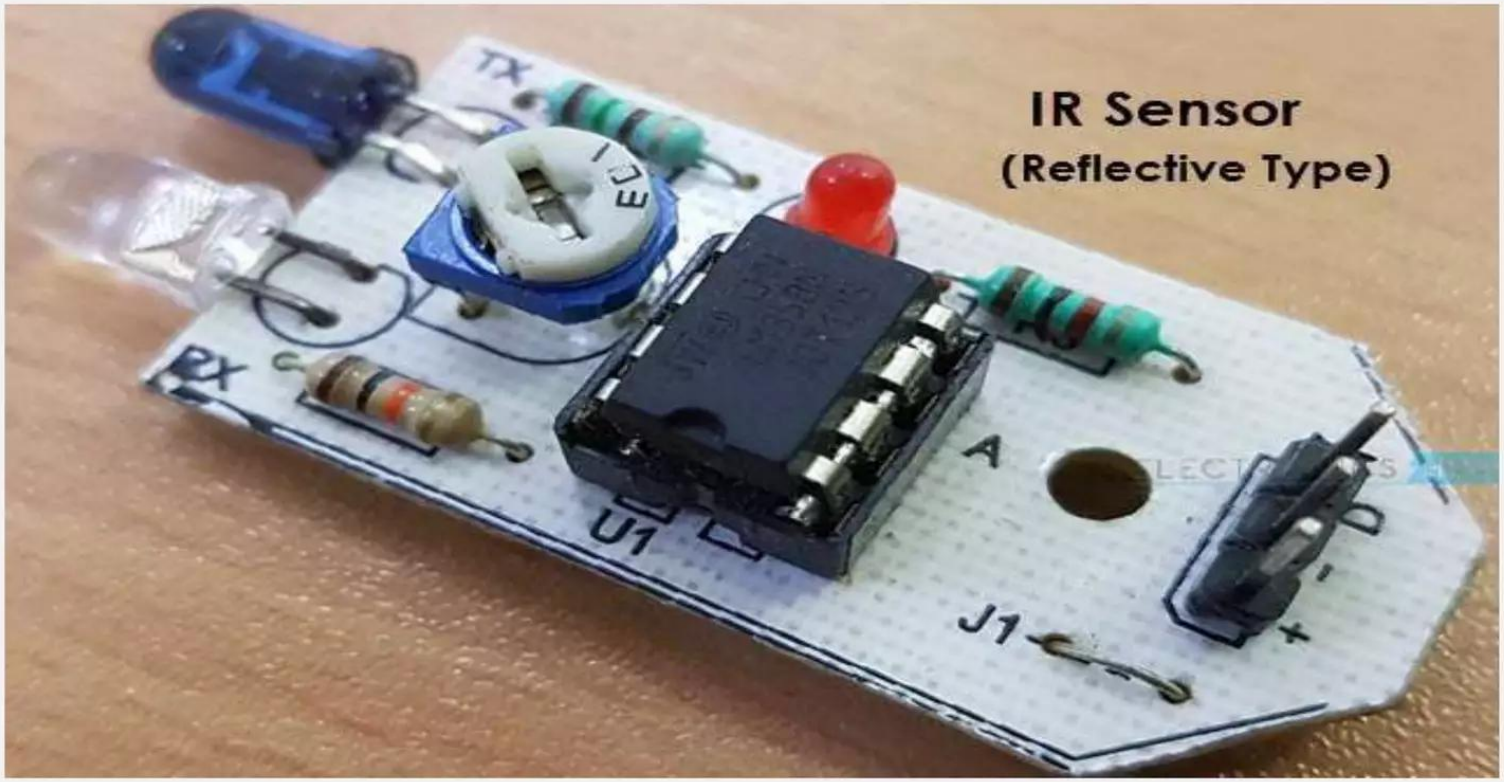
- A Proximity Sensor is a non-contact type sensor that detects the presence of an object. Proximity Sensors can be implemented using different techniques like Optical (like Infrared or Laser), Ultrasonic, Hall Effect, Capacitive, etc.

•Some of the applications of Proximity Sensors are Mobile Phones, Cars (Parking Sensors), industries (object alignment), Ground Proximity in aircraft, etc.



INFRARED SENSOR (IR SENSOR)

- IR Sensors or Infrared Sensor are light based sensor that are used in various applications like Proximity and Object Detection. IR Sensors are used as proximity sensors in almost all mobile phones.
- There are two types of Infrared or IR Sensors: **Transmissive Type and Reflective Type.**
- In **Transmissive Type IR Sensor**, the IR Transmitter (usually an **IR LED**) and the **IR** Detector (usually a Photo Diode) are positioned facing each other so that when an object passes between them, the sensor detects the object.
- The other type of IR Sensor is a **Reflective Type IR Sensor**. In this, the transmitter and the detector are positioned adjacent to each other facing the object. When an object comes in front of the sensor, the sensor detects the object.
- Different applications where IR Sensor is implemented are Mobile Phones, Robots, Industrial assembly, automobiles etc.

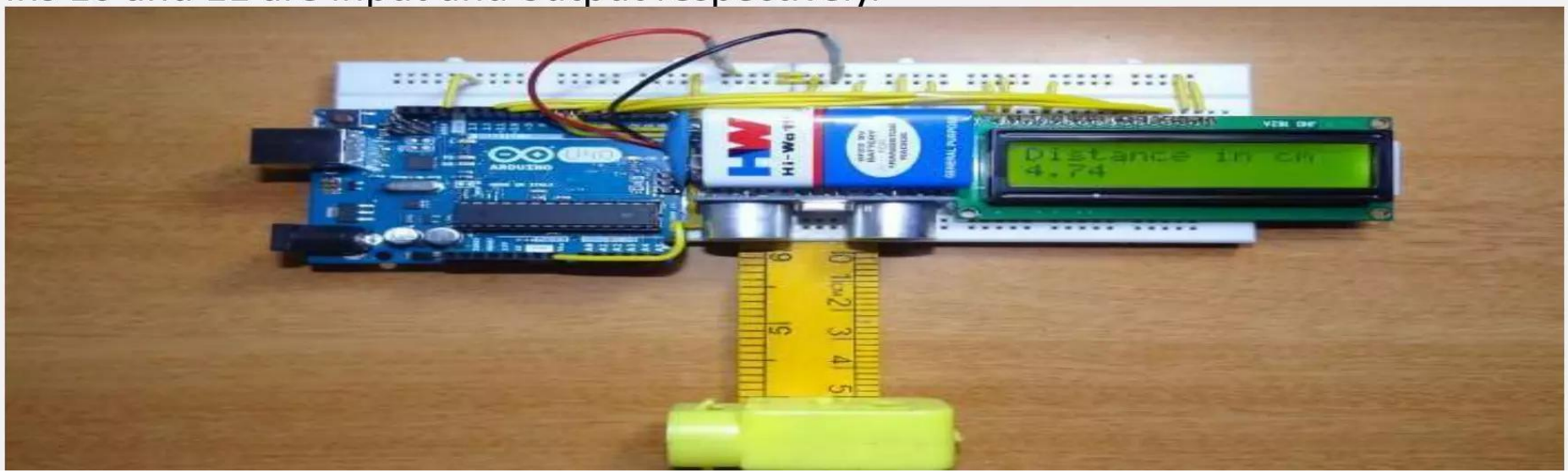

IR Sensor
(Reflective Type)
TX
RX
U1
J1
A
+
-

ULTRASONIC SENSOR

- An Ultrasonic Sensor is a non-contact type device that can be used to measure the distance as well as the velocity of an object. An Ultrasonic Sensor works based on the properties of the sound waves with a frequency greater than that of the human audible range.
- There are two kinds of ultrasonic sensors, “**active ultrasonic sensors**” and “**passive ultrasonic sensors”**. An active ultrasonic sensor generates the high-frequency sound wave to receive back the ultrasonic sensor for evaluating the echo. But, passive ultrasonic sensors are just used for detecting ultrasonic noise.



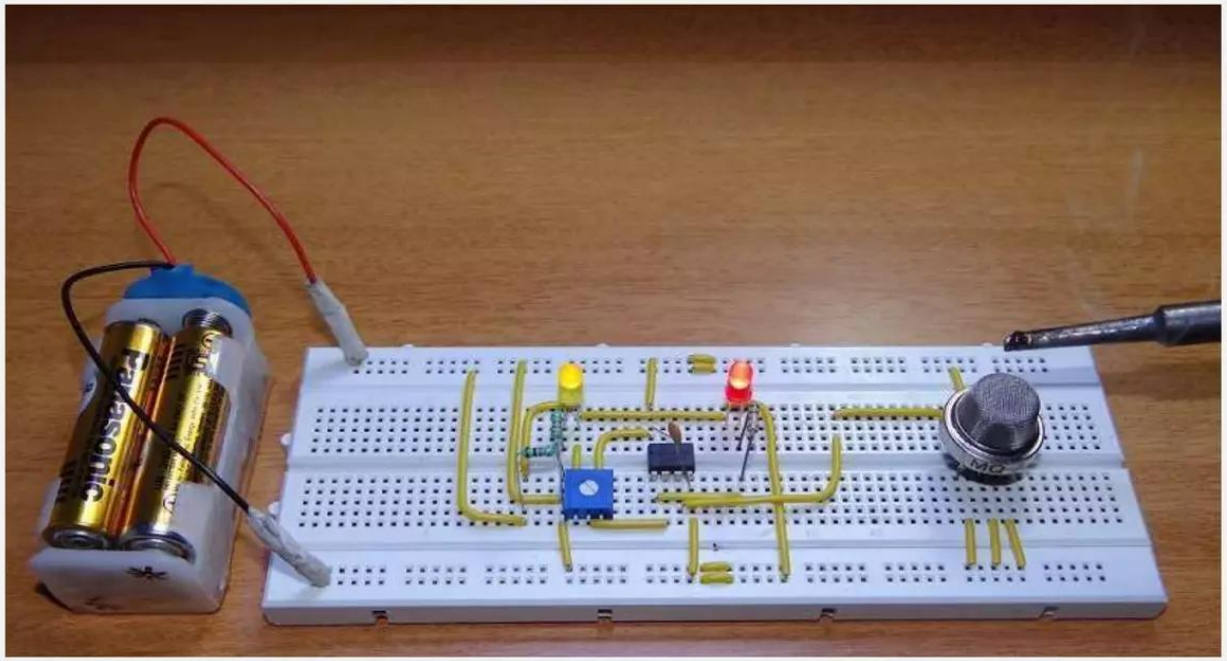
- Ultrasonic means nothing but the range of the frequencies. Its range is greater than the audible range (>20 kHz).
- This ultrasonic sensor is ***used to calculate the distance between the ultrasonic transmitter and the target and also used to measure the velocity of the target*.**
- **Ultrasonic sensor HC-SR04** can be used to measure distance in the range of 2cm-400cm with an accuracy of 3mm.
- The 4 pins of the Ultrasonic sensor are Vcc, Gnd, Trig and Echo. Trig is connected to Pin 11 of Arduino and Echo is connected to Pin 10. With respect to Arduino, Pins 10 and 11 are input and output respectively.



CHEMICAL/SMOKE AND GAS SENSOR

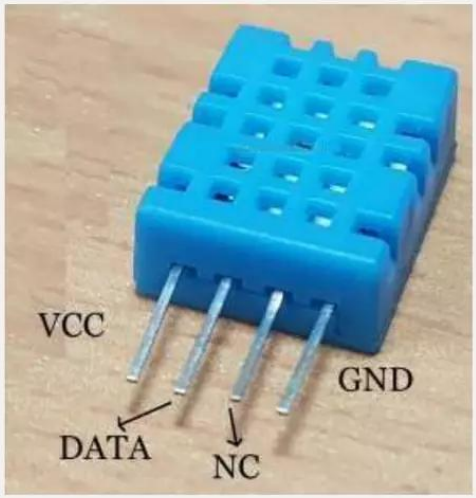
- smoke and gas detector is a gadget that sense gas, smoke and typically it's an indicator of fire. Now a days all security devices using this sensor to passing signal to fire alarm to control panel. Household smoke detector is also known as smoke alarm, most of the device manufacturer using audible or visual alarm system in security devices that detect automatically.





HUMIDITY SENSOR

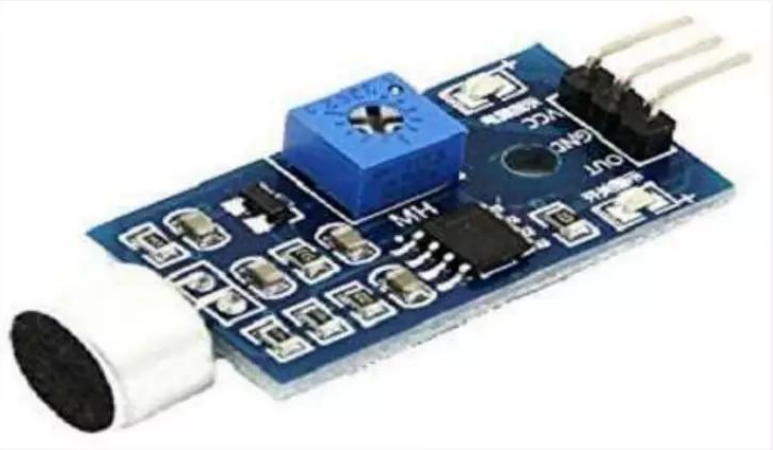
- A humidity sensor (or hygrometer) senses, measures and reports both **moisture and air temperature**. The ratio of moisture in the air to the highest amount of moisture at a particular air temperature is called **relative humidity(RH)**. Relative humidity becomes an important factor when looking for comfort.





SOUND SENSOR

- The sound sensor is one type of module used to notice the sound. Generally, this module is used to detect the intensity of sound.
- This sensor employs a microphone to provide input to a buffer, peak detector and an amplifier. This sensor notices a sound, & processes an o/p voltage signal to a microcontroller. After that, it executes the required processing.
- Pin1 (VCC): 3.3V DC to 5V DC
- Pin2 (GND): This is a ground pin
- Pin3 (DO): This is an output pin



THANK YOU!!